VEER NARMAD SOUTH GUJARAT UNIVERSITY

University Campus, Udhna-Magdalla Road, Surat – 395001, Gujarat, India.

Department of Information and Communication Technology

M.Sc.(ICT) Program

PROJECT TITLE

“ SVM Education System”

AS PARTIAL REQUIREMENT FOR THE DEGREE

OF

MASTER OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (MSC ICT 2 Year Course)

2025-26 (2nd Semester)

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Re-Accredited 'B++' 2.86 CGPA by NAAC
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M.Sc.(Information and Communication Technology) Programme

Certificate

This is to certify that **Ms. Prakruti Pareshbhai Chapatwala**, with Exam **Seat Number 18** and **Enrollment/Registration ID R25110018000710006**, has worked on her project entitled **SVM Education System** at the **Department of ICT** as a partial fulfillment of the requirement for the degree of M.Sc. (Information and Communication Technology) – 2nd Semester, during the academic year 2025–2026.

Date:21/06/2026
Place:Department of ICT, VNSGU, Surat

Internal Project Guide	Course Coordinator	Head of the Department
M.Sc.(I.C.T.) 2nd Semester	M.Sc. (I.C.T.) Programme	Department of I.C.T.
Department of I.C.T.	Department of I.C.T.	Veer Narmad South Gujarat University, Surat
Veer Narmad South Gujarat University, Surat	Veer Narmad South Gujarat University, Surat	

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Department Of Information And Communication Technology

M.Sc.(Information and Communication Technology) Programme

Certificate

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Date:21/06/2026
Place:Department of ICT, VNSGU, Surat

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ACKNOWLEDGEMENT

When we reach the completion of the project, giving credit becomes a must, as without the support of other so many people's help and guidance, this project couldn't be completed successfully.

Success in such a comprehensive project cannot be achieved single-handled. It is team effort that sails the ship to the coast. So, we would like to express our sincere thanks to all the dignitaries who were involved in making this project a great joy and turning it into successful piece of work.

We would like to take opportunity to thank our Department J.P. Dawer Institute of Information Science & Technology, Surat for giving us this tremendous opportunity to work in the real-time project. We are also thankful to Sharda Vidyamandir, managed by Shivshakti Education Trust, for allowing us to carry out this project and for providing us with the necessary data, information, and support required for analysis and successful completion of the project.

Ms.Nisha rana , our professor and project coordinator, has been very prudent to us throughout our college studies. She is the person who has been giving direction to our work and the shape to our imagination and always ready to give best guidance, give solutions whenever required. We express our regards to her from the core of our heart.

Last but not the least, our heartfelt appreciation goes to all those not named here, but who have rendered their co-operation, little or more, directly or indirectly involved in the development of this system.

Thanking All.
Prakruti Pareshbhai Chapatwala ,
Bhagvat Rahul KailashBhai

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1.Introduction

1.1 Organization Profile

Sharda Vidyamandir is a well-established educational institution located at Opposite Gujarat Housing Board, Gopal Nagar, Pandesara, Surat-394221, Gujarat. The school provides quality education in Gujarati, Hindi, and English mediums, catering to students from primary to higher secondary levels.

The institution focuses on overall development of students by maintaining academic excellence along with extracurricular activities. It manages various records such as student information, attendance, examination details, and administrative data in an organized manner. With the growing need for digitalization, the school requires an efficient system to manage its operations effectively and accurately.

1.2 User/Client Profile

The main users of the system are school administrators, staff members, and teachers. These users are responsible for managing student records, maintaining data, and handling various academic and administrative activities. The system is designed to be user-friendly so that authorized users can easily perform their tasks efficiently.

1.2.1 Current System

Currently, the school uses a combination of manual methods and a third-party software system (VMS) to manage its operations. Many records such as student details, attendance, and administrative data are maintained manually in registers. Along with this, some data is handled using the VMS software.

However, the existing system has several limitations. The manual process is time-consuming, requires significant effort, and is prone to human errors. On the other hand, the VMS software, being a third-party system, does not fully meet the specific requirements of the school. It lacks customization, flexibility, and complete control over data management.

Due to these issues, it becomes difficult to manage data efficiently, and there may be chances of data inconsistency and redundancy. Therefore, there is a need for a dedicated system that is specifically designed according to the requirements of the school to overcome these challenges.

1.2.2 System Details

The proposed system is designed specifically for Sharda Vidyamandir to overcome the limitations of the existing manual and third-party (VMS) system. It will provide a centralized and efficient platform to manage all school-related data.

The system will allow administrators and staff to store, update, and manage student records, attendance, and other important information digitally. It will simplify daily administrative tasks and improve the overall working process of the school.

In addition to the admin side, the system will also include a user-side web interface (school webpage). Through this, users such as students and parents can access general information about the school, view notices, and check relevant details online. This will improve communication and make information easily accessible.

Unlike the current system, the proposed system is fully customizable according to the school's specific requirements. It will reduce dependency on manual work and third-party software, minimize errors, and provide quick access to data.

Overall, the system will ensure better data security, consistency, and reliability, while improving the efficiency and transparency of school management.

2. Proposed System

2.1 Scope

The scope of the proposed system is to manage and maintain all school-related data efficiently in a centralized manner. The system will be used by administrators, staff, teachers, and students for performing various academic and administrative tasks.

It will handle student records, teacher details, staff information, and other important data digitally. The system will also include an online school webpage through which users can communicate, view school details, staff information, and photographs.

Additionally, it will provide features such as admission inquiries and easy access to information for users. Overall, the system aims to simplify school operations and improve communication between the school and users.

2.2 Objectives

The objective of the School Management System is to provide an efficient, reliable, and user-friendly digital platform for managing all school-related activities. The system is designed to simplify administrative work, improve data management, and enhance communication between the school, students, and parents. It aims to replace manual and third-party systems with a customized solution tailored to the specific needs of the school.

1. Manage School Data Efficiently
To maintain student, teacher, and staff records in a structured and centralized database for easy access and management.

2. Reduce Manual Work and Errors
To minimize the use of manual registers and reduce human errors by digitizing all important records and processes.

3. Provide Centralized Data Management
To store all school-related information in one system, ensuring better organization, consistency, and data security.

4. Simplify Administrative Tasks
To provide an easy-to-use system for administrators and staff to manage daily operations such as updating records and handling information.

5. Develop an Informative School Webpage
To create a user-friendly website where users can view school details, staff information, and other important updates.

6. Improve Communication with Users
To allow effective communication between the school and users through the website for notices, updates, and inquiries.

7. Enable Online Admission Inquiry
To provide a feature for users to submit admission inquiries online, making the process easier and faster.

8. Provide Easy Access to Information
To allow students, parents, and other users to access relevant school information anytime through the web interface.

9. Improve Overall Efficiency of School Management
To enhance the speed, accuracy, and reliability of managing school operations through a digital system.

2.3 Constraints

2.3.1 Hardware Constraints

• The performance of the system depends on internet connectivity. Slow or unstable network may affect the usage of the web application on both computers and mobile devices.

• As the system is a web application, it can be accessed through devices such as desktops, laptops, and smartphones. However, proper device compatibility and minimum hardware specifications are required for smooth functioning.

• Limited availability of computers in the school may restrict access for staff, although mobile devices can be used as an alternative.

• Adequate storage and memory are required on the server side to store and manage school data efficiently.

2.3.2 Software Constraints

• Requires .NET Core supported environment

• Requires properly configured SQL database

• Needs updated and compatible web browser

• Internet connection is required

• May not work on outdated software systems

2.4 Advantages

• Efficient Data Management
The system allows administrators and staff to manage student, teacher, and staff records efficiently. All data is stored in a structured manner, reducing manual work and improving organization.

• Reduced Manual Work
The system eliminates the need for maintaining records in registers. It reduces paperwork, saves time, and minimizes human errors in data handling.

• Centralized System
All school-related data is stored in a single system, making it easy to access, update, and manage information whenever required.

• Improved Communication
The school webpage allows users to view important information, notices, and updates, improving communication between the school and users.

• User-Friendly Interface
The system provides a simple and easy-to-use interface for administrators, staff, students, and parents, making operations smooth and efficient.

• Web-Based Accessibility
The system can be accessed from anywhere using internet-enabled devices such as computers and mobile phones, increasing flexibility and usability.

• Data Security and Reliability
All data is stored securely in the database, reducing the risk of data loss and ensuring reliable data management.

2.5 Limitations

• Requires Internet Connectivity
The system depends on internet access. Poor or unstable connection may affect system performance.

• **Basic Technical Knowledge Required**
Users must have basic knowledge of using computers or mobile devices to operate the system effectively.

• **Device and Browser Compatibility**
Older devices or outdated browsers may not support all features of the system properly.

• **Initial Setup and Maintenance**
The system requires proper setup, configuration, and regular maintenance for smooth functioning.

• **Limited Features Compared to Large Systems**
The system is developed for a specific school, so it may not include all advanced features available in large commercial software.

• **Server Dependency**
If the server is down or not functioning properly, the system will not be accessible.

3.Environment Specification

3.1 Hardware and Software Requirements

- **Hardware Requirements**

Processor	AMD Ryzen 3 5300U with Radeon Graphics 2.60 GHz
RAM	**8.00 GB**
Disk Space	**475 GB of Available SSD**
Graphic	**AMD Ryzen 3 5300U with Radeon Graphics**
Display	**1920*1080**

- **Software Requirements**

Component Type	Technology / Tool Used
Operating System	Windows 11
Front End	ASP.NET MVC Core (C#)
Back End	C# (.NET Core Framework)
Code Editor	Visual Studio
Database	SQL Server / MySQL
Local Server	IIS (Internet Information Services) – used for running and testing application locally
Web Browser	Google Chrome
Drawing Tools	Draw.io

3.2 Development Description

The project is developed using **ASP.NET MVC Core with C#**, following the MVC (Model-View-Controller) architecture. This architecture helps in separating the application into three main components: Model, View, and Controller, making the project more organized and easier to maintain.

The frontend is designed using **Razor Views (HTML, CSS, and Bootstrap)** to provide a clean and user-friendly interface. The backend logic is implemented in **C# (.NET Core Framework)** to handle all business operations and application functionality.

SQL Server / MySQL database is used to store and manage all application data securely and efficiently. The system supports **CRUD operations (Create, Read, Update, Delete)** for managing records.

The application is developed and tested using **Visual Studio** as the code editor and run on **IIS (Local Server)**. The system is tested in **Google Chrome browser** to ensure proper performance and compatibility.

4.System Planning

4.1 Feasibility Study

Feasibility study is an important phase of System Planning which is used to determine whether the project is practically possible to develop within the given constraints of time, cost, and technology.

1. Technical Feasibility

The project is technically feasible because it is developed using **ASP.NET MVC Core (C#)** and **SQL Server / MySQL**, which are stable, secure, and widely used technologies. Visual Studio provides a powerful development environment for coding, debugging, and testing.

2. Economic Feasibility

The project is economically feasible because it uses freely available tools like Visual Studio Community Edition, SQL Server Express/MySQL, and Windows OS. Therefore, no additional high cost is required for development.

3. Operational Feasibility

The system is user-friendly and easy to operate. It provides a simple interface for users, making it easy to perform operations like login, data entry, and record management without technical difficulty.

4.2 Software Engineering Model

The project is developed using the **MVC (Model-View-Controller) architecture**, which is a widely used software design pattern in ASP.NET applications.

Model

- Represents the data structure of the application
- Handles database operations using Entity Framework / SQL queries
- Responsible for business logic and data validation

View

- Represents the user interface of the application
- Developed using Razor Pages, HTML, CSS, and Bootstrap
- Displays data to the user in a structured format

Controller

- Acts as an interface between Model and View
- Handles user requests and application logic
- Sends and receives data from Model and updates View accordingly

This architecture improves **code reusability, scalability, and maintainability** of the system.

4.3 Risk Analysis

Risk analysis identifies potential issues that may affect the development, deployment, or operation of the system. Proper strategies are defined to minimize and manage these risks effectively.

1. Technical Risks

• **Server Downtime:** IIS server may crash or stop temporarily, affecting system availability.
• **Database Connection Failure:** SQL Server / MySQL issues may prevent data storage or retrieval.
• **Browser Compatibility:** Minor UI issues may occur on different browsers or older versions.
• **Performance Issues:** High number of users or large data may slow down system response time.

2. Security Risks

• **Unauthorized Access:** Weak authentication may allow unauthorized users to access sensitive data.
• **Data Loss / Corruption:** Improper backup or system failure may result in loss of important records.
• **Cyber Attacks:** Risks such as SQL injection, hacking attempts, or data manipulation may affect system integrity.

3. Operational Risks

• **User Resistance:** Some users may find it difficult to adapt to the new system initially.
• **Incorrect Data Entry:** Manual mistakes during data entry may lead to incorrect records.
• **Training Requirements:** Administrators may require basic training to operate the system effectively.

4. Environmental / External Risks

• **Network Issues:** Slow or unstable internet connection may affect system usage.
• **Power Failure:** Sudden power cuts may interrupt system operations or data processing.
• **System Updates:** Changes in technology or software versions may require future updates.

5. Project Management Risks

• **Time Delays:** Unexpected errors or issues may delay project completion.
• **Scope Creep:** Addition of extra features during development may increase workload.
• **Resource Limitations:** Limited tools or manpower may affect development speed and quality.

4.4 Project Schedule

Project schedule defines the overall planning of activities required for the completion of the system within a defined time period. It helps in organizing tasks in a sequential manner and ensures timely delivery of the project.

The project is divided into different phases such as requirement analysis, system design, development, testing, and deployment. Each phase is assigned a specific time duration to maintain proper workflow.

4.4.1 Task Dependency

Dependency	Description
Requirement Analysis → System Design	System design (architecture, DFD, ERD) is prepared after requirement study.
System Design → Database Creation	Database structure and relationships are created based on system design.
Database Creation → API Development	APIs are developed after database schema is finalized.
API Development → Backend Development	Backend logic uses APIs for data processing and business operations.
Backend Development → Frontend Development	Frontend UI depends on APIs for data fetching and operations.

Frontend Development → Module Integration	All frontend and backend modules are integrated into the system.
Module Integration → Testing	Testing is performed after full integration of system modules.
Testing → Deployment	Final deployment is done after successful testing and bug fixing.

4.4.2 Timeline Chart

4.4.3 Project Table

Phase	Duration	Description
Requirement Analysis	2 Weeks	System requirements collection and feasibility study
System Design	2 Weeks	Architecture, DFD, and database design preparation
Database Design	1 Week	Tables, relationships and schema creation
API Development	2 Weeks	REST APIs creation for data communication
Backend Development	3 Weeks	Business logic implementation using ASP.NET Core (C#)
Frontend Development		UI development using Razor Views, HTML, CSS
Module Integration		
Testing Phase		
Deployment		

5 System Analysis

5.1 Detailed SRS (System Requirements Specification)

1) User Module

1 User Registration & Authentication

- **Description:**
  This module allows users to create an account and login into the system using valid credentials stored in the `users` table.
- **Input:**
  username, password, full_name, email, phone, group_id
- **Events:**
  User registers → data stored in `users` table → login authentication performed
- **Output:**
  Successful login / error message
- **Validations:**
  - Username must be unique
  - Email must be valid format
  - Password must not be empty
- **Constraints:**
  - Only registered users can access system
  - Password stored securely

2 Student Management (Profile View)

- **Description:**
  Displays student details from `students` table.
- **Input:**
  student_id / user_id
- **Events:**
  User requests profile → data fetched from database
- **Output:**
  Student profile details
- **Validations:**
  - Valid student_id required
- **Constraints:**
  - Users can only view their own data

3 Admission Inquiry Module

- **Description:**
  Allows users to submit admission inquiries stored in `admission_inquiry` table.
- **Input:**
  student_name, parent_name, phone, class_name, message
- **Events:**
  Inquiry form submitted → stored in database

- **Output:**
  Inquiry confirmation message
- **Validations:**
  - Phone number must be valid
  - Required fields must not be empty
- **Constraints:**
  - Inquiry date auto-generated

4 Fee Payment Module

- **Description:**
  Handles student fee payments using `fee_payment` and `fee_structure` tables.
- **Input:**
  student_id, fee_id, amount_paid, payment_mode
- **Events:**
  Payment initiated → stored in database → receipt generated
- **Output:**
  Payment confirmation & receipt
- **Validations:**
  - Amount must be valid
  - Student must exist
- **Constraints:**
  - Payment linked with fee_structure

2) Admin Module

1 User Management

- **Description:**
  Admin manages all system users from `users` table.
- **Input:**
  user details
- **Events:**
  Add / update / delete users
- **Output:**
  Updated user list
- **Validations:**
  - Duplicate username not allowed
- **Constraints:**
  - Only admin has access

2 Student Management

- **Description:**
  Admin manages student records in `students` table.
- **Input:**
  student details
- **Events:**
  Add / update / delete student records

- **Output:**
  Student database updated
- **Constraints:**
  - Linked with classes, sections, sessions

3 Class, Section & Subject Management

- **Description:**
  Manages academic structure using:
  - `classes`
  - `sections`
  - `subjects`
- **Input:**
  class_name, section_name, subject_name
- **Events:**
  Data entry → stored in database → linked relations created
- **Output:**
  Structured academic data
- **Constraints:**
  - Foreign key relationships must be maintained

4 Staff Management

- **Description:**
  Handles staff details using `staff` table.
- **Input:**
  name, designation, salary, qualification, phone
- **Events:**
  Staff record management by admin
- **Output:**
  Staff data list
- **Constraints:**
  - Linked with users table

5 Attendance Module

- **Description:**
  Manages attendance of students and staff.
- **Tables Used:**
  - `student_attendance`
  - `staff_attendance`
- **Input:**
  attendance date, status
- **Output:**
  Attendance reports
- **Constraints:**
  - One entry per date per user

6 Fee Structure Management

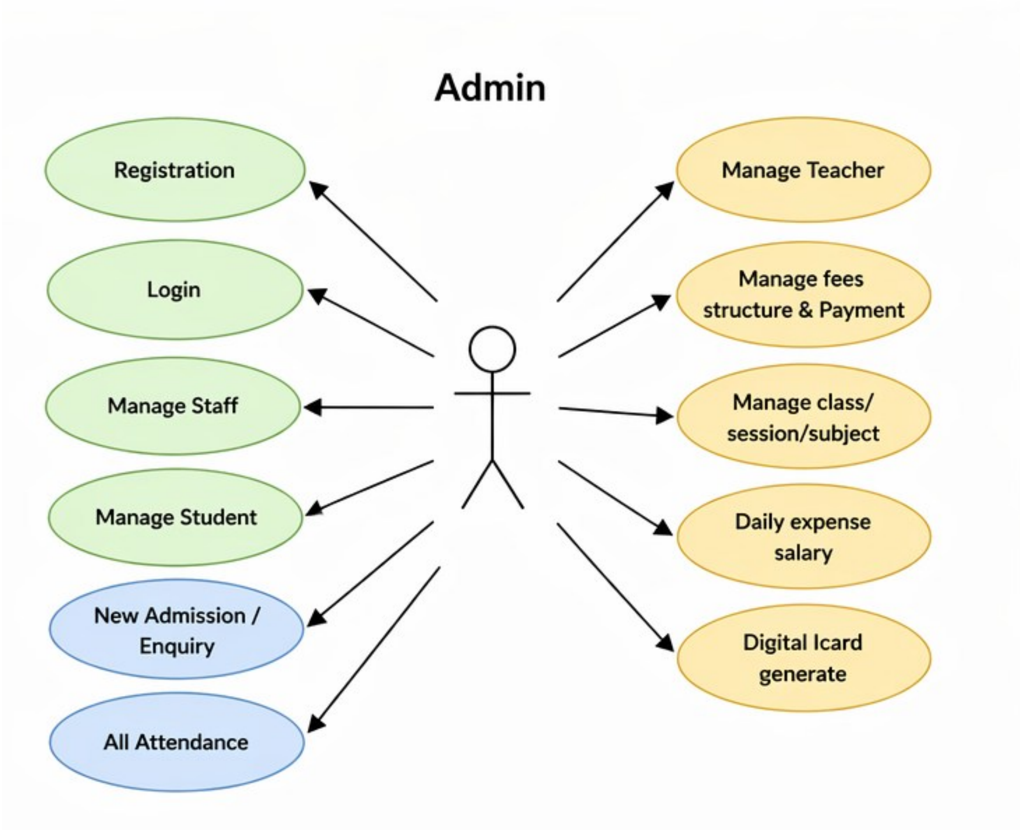
- **Description:**
  Admin manages fee structure using `fee_structure` table.
- **Input:**
  class_id, fee_type, amount, due_date
- **Output:**
  Fee structure list
- **Constraints:**
  - Linked with class and section

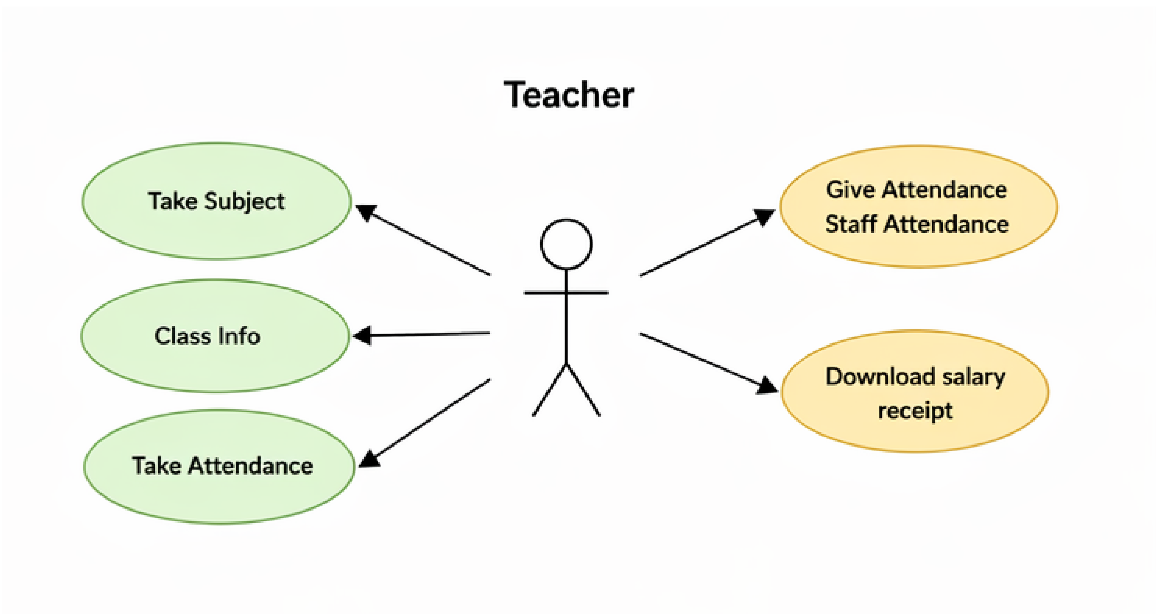
7 Sessions Management

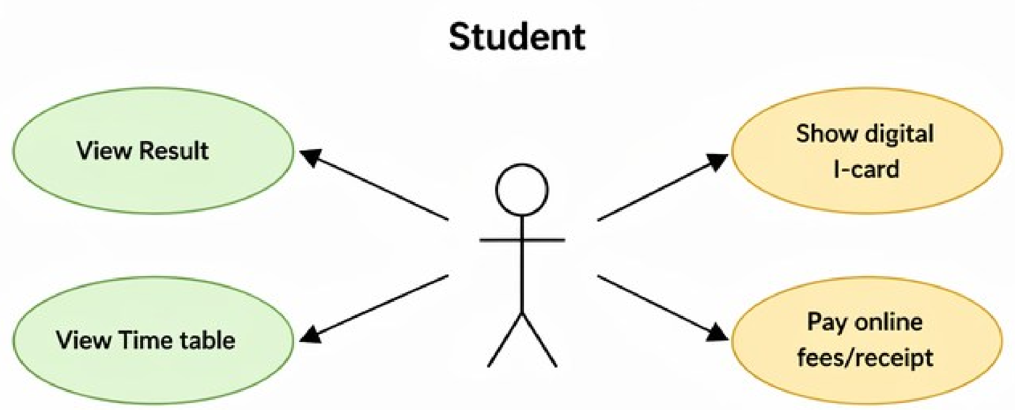
- **Description:**
  Handles academic sessions using `sessions` table.
- **Input:**
  session_name, start_date, end_date
- **Output:**
  Active session data
- **Constraints:**
  - Only one active session allowed

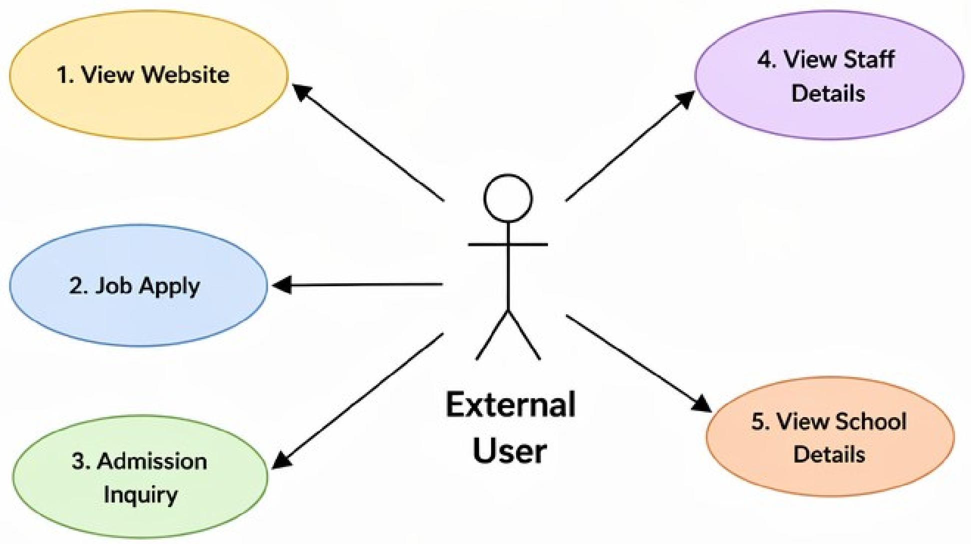
5.2 UML Diagram

5.2.1 Use Case Diagram

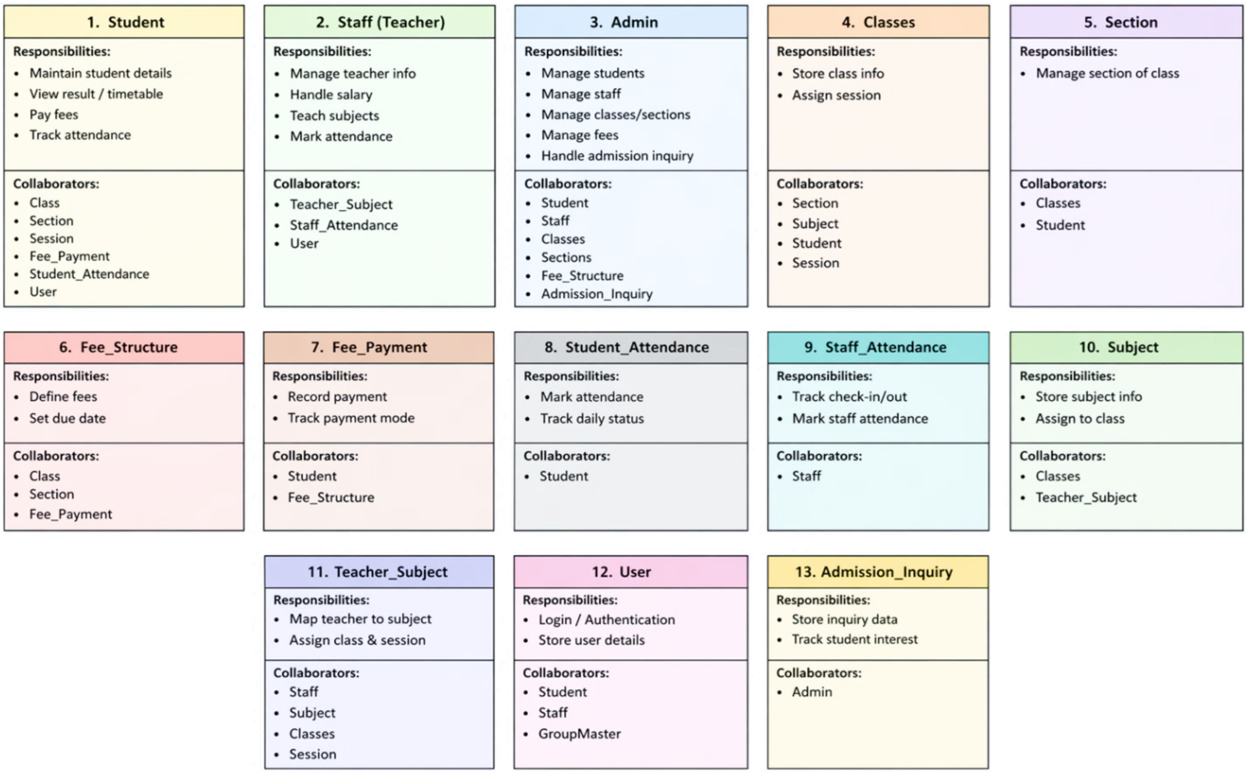




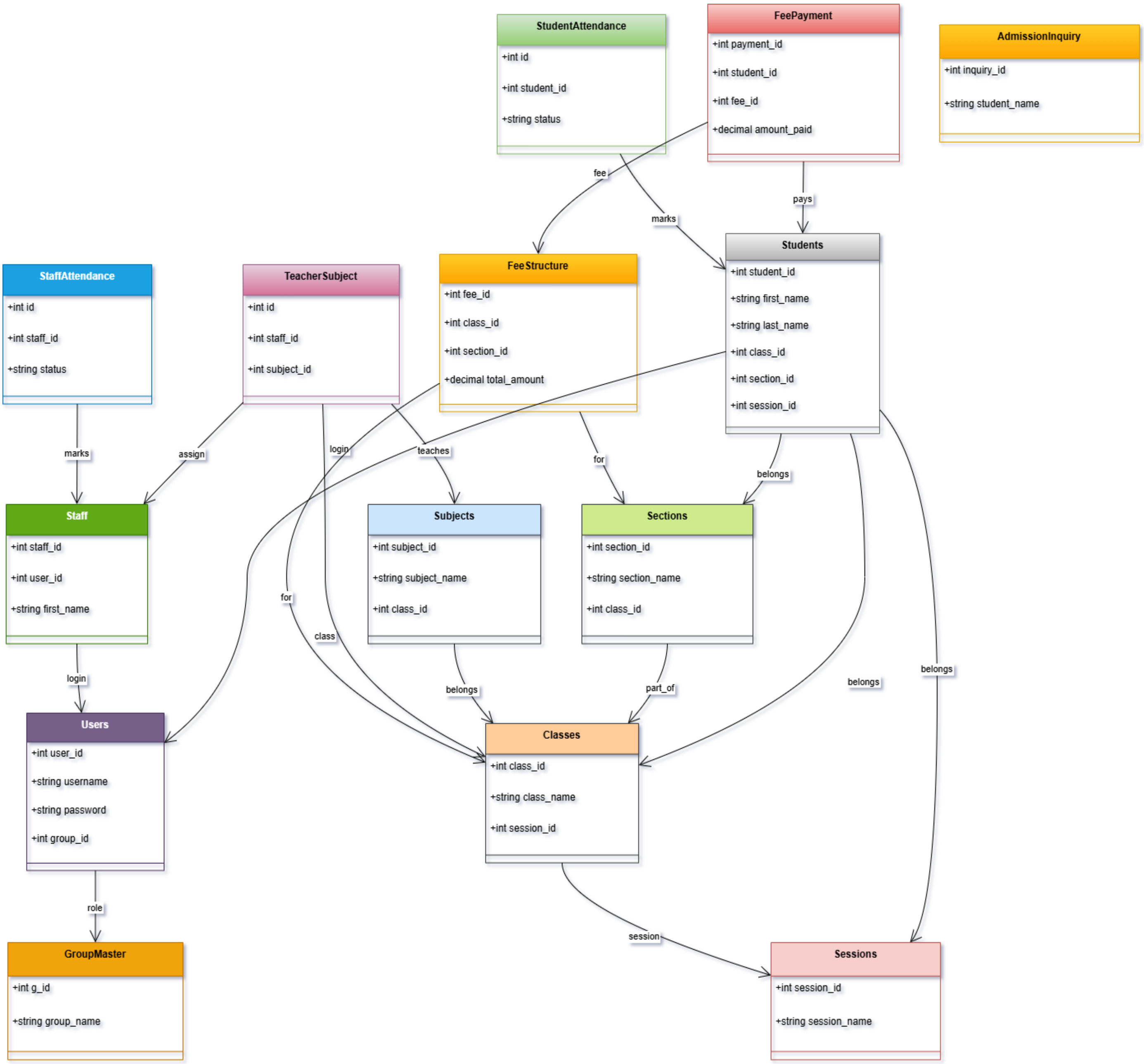

Student
View Result
View Time table
Show digital I-card
Pay online fees/receipt


1. View Website
2. Job Apply
3. Admission Inquiry
External User
4. View Staff Details
5. View School Details

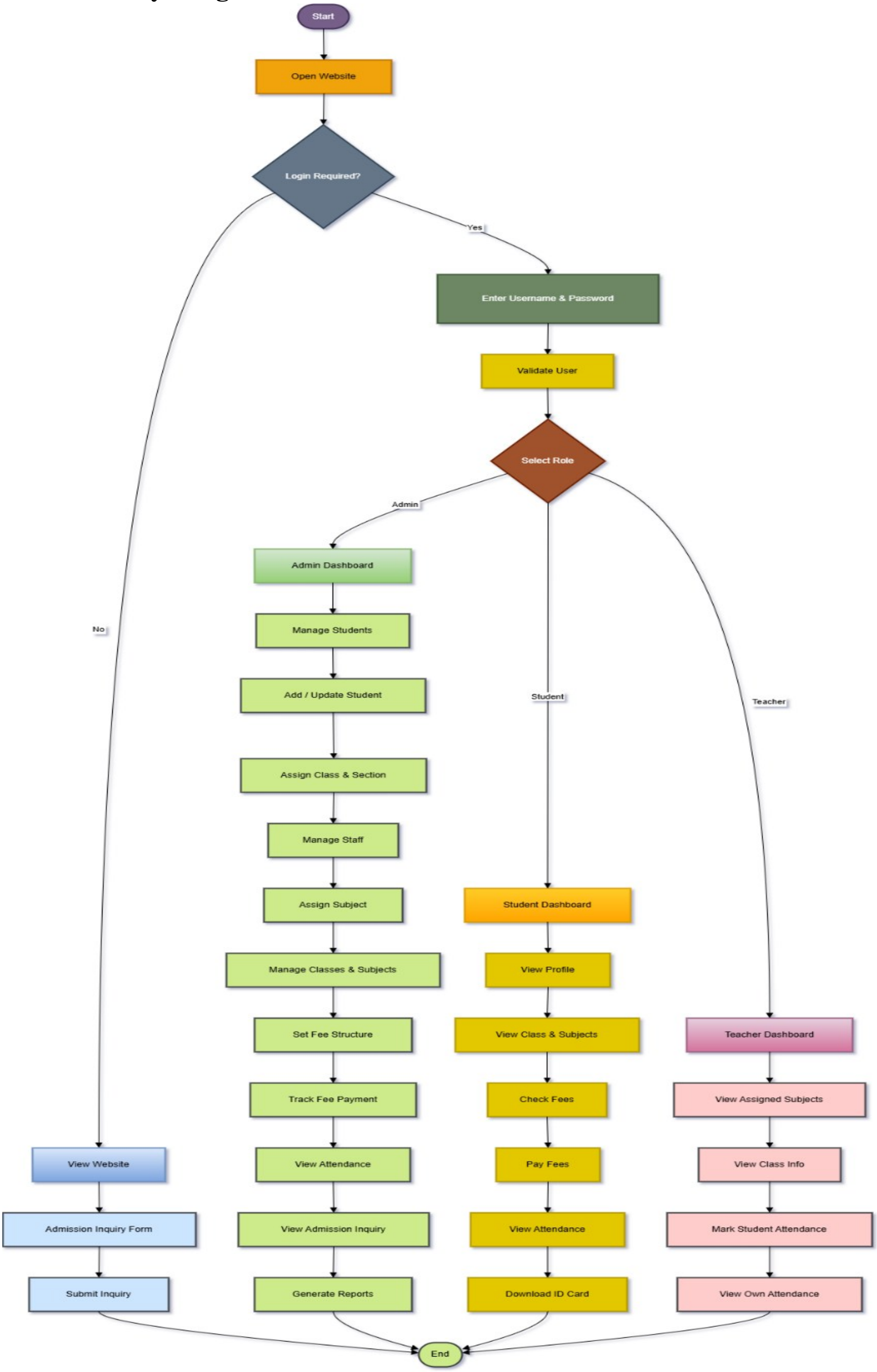
5.2.2 **CRC Diagram**



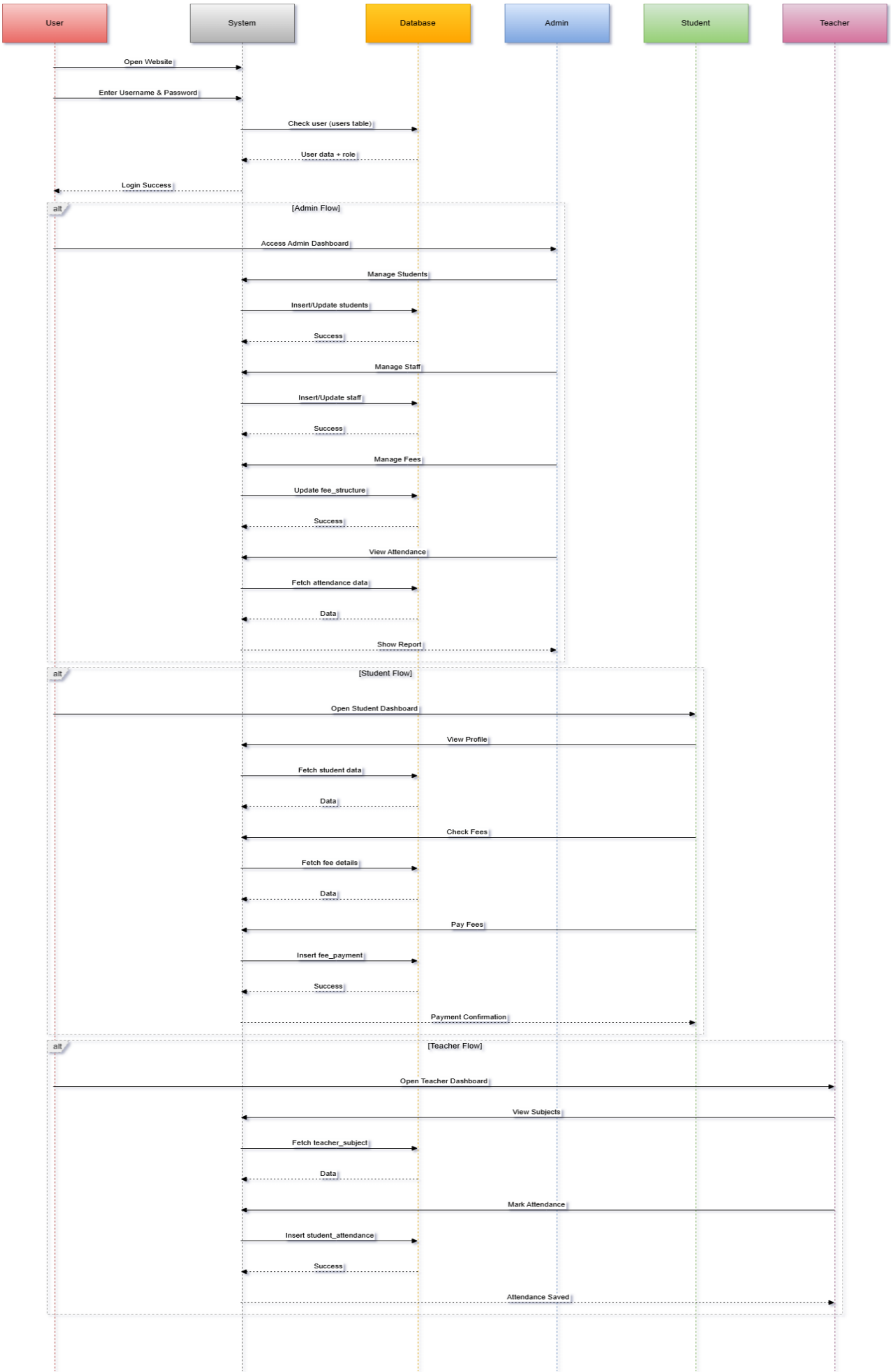
5.2.3 Class Diagram



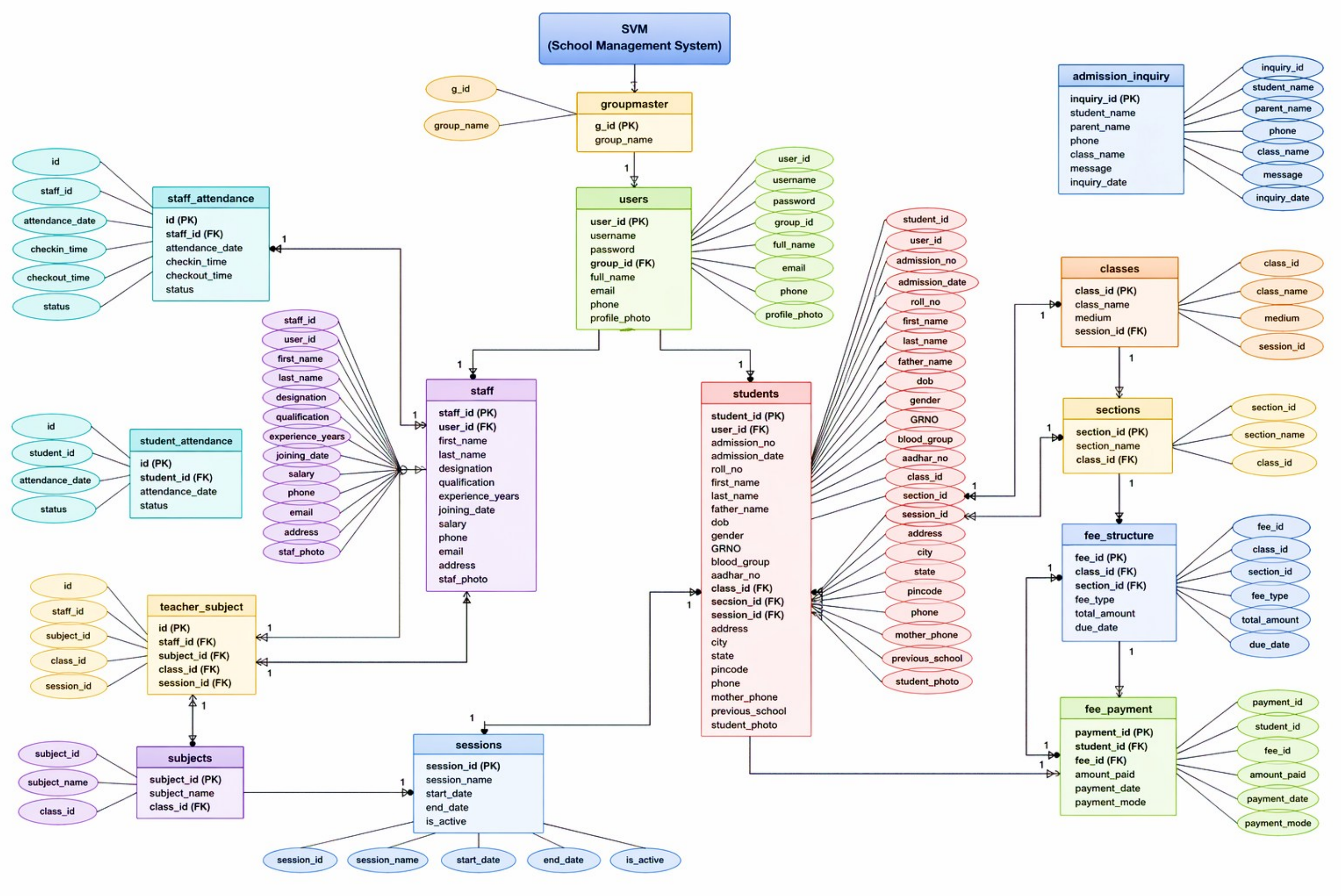
5.2.4 Activity Diagram



5.2.5 Sequence Diagram



5.3 E-R Diagram



6.Software Design

6.1 Database Design